

AeroMorse Switch Mode Guide — Two-Button Hold Mode

Switch mode (Group 7, new in v1.15) turns AeroMorse into **two plain switches**. There is no Morse code in this group:

- a **sip** presses **Enter** the moment it starts and **holds it** until the sip ends;
- a **puff** presses **Space** the moment it starts and **holds it** until the puff ends.

That is exactly how a pair of physical accessibility switches behaves, so it works with games and apps built for one or two switches — including ones that need a key **held down**, which Morse groups can't do.

When to use which group

	Group 4 — Scanning	Group 7 — Switch
What a sip/puff does	Taps a key after the Morse pattern finishes	Presses a key instantly and holds it while you sip/puff
Keys	12: Enter, Space, F3–F12 (1–3-symbol patterns)	2: Enter (sip), Space (puff)
Holding a key down	No	Yes
Best for	iOS Switch Control, Android Switch Access, Samsung Universal Switch	Switch games and apps, anything that needs "hold Space"
Guide	AEROMORSE_SWITCH_CONTROL_GUIDE.md	this guide

Quick start

1. **Enter Switch mode:** type the Group 0 code `...-----` (3 sips, 5 puffs). The screen shows **[SWITCH]**. (Long-press group cycling skips Switch mode, so you never land in it by accident.)
2. **Sip** to hold **Enter**, **puff** to hold **Space**. While a key is held the screen shows `HOLD RETURN` or `HOLD SPACEBAR`.
3. **Leave Switch mode:** just **stop** — after **20 seconds** with no sip or puff it goes back to **Keyboard** by itself. (Or hold one puff for **5 seconds**: Space is let go at 5 s and you switch to Keyboard.)

Things that do **not** change group in Switch mode, on purpose:

- **Hard (strong) sips and puffs** — games often need them.

- **Long sips and puffs** — games often need long holds.
- **Quick repeated sips or puffs.**

The first sip or puff after AeroMorse starts only closes the start-up (version) screen; it doesn't press a key.

Benny's Hub

Benny's Hub (by NARBE) is a free library of games and tools made to be used with **one or two buttons mapped to Space and Return** — a good match for Switch mode, and it runs in any web browser on the computer AeroMorse is plugged into.

How the hub is driven:

AeroMorse (Switch mode)	Key	In Benny's Hub
Puff	Space	Move to the next item
Sip	Enter (Return)	Choose / open the highlighted item

Games that need a held key. Some games use how long you hold. In **NARBE Kart** (the racing game), for example, the on-screen help says *Hold Space to slide left, hold Enter to slide right*, and its menus use *tap Space = next, hold Space = back, Enter = choose*. In Switch mode that is simply:

- **hold a puff** → hold Space → slide left;
- **hold a sip** → hold Enter → slide right;
- short puff → next; long puff → back; sip → choose.

Getting started:

1. On the computer, open <https://narbehouse.github.io/bennysub/index.html> in a browser (full screen helps).
2. On AeroMorse, enter Switch mode (`...-----`).
3. Puff to move, sip to choose. When you're finished, stop for 20 seconds and AeroMorse returns to Keyboard.

Some Benny's Hub games also have a **one-switch** option. In Switch mode you can use just the puff (Space) for those.

Settings (`config.py`)

All in the **SWITCH GROUP** section of `config.py` . Save and the device restarts with the new values.

Setting	Default	What it does
<code>SWITCH_GROUP</code>	<code>7</code>	Which group is Switch mode. <code>0</code> = no Switch mode (Group 7 becomes an ordinary placeholder group again).
<code>SWITCH_SIP_KEY</code>	<code>"ENTER"</code>	Key held while sipping. Any Keycode name, e.g. <code>"LEFT_ARROW"</code> .
<code>SWITCH_PUFF_KEY</code>	<code>"SPACE"</code>	Key held while puffing, e.g. <code>"RIGHT_ARROW"</code> .
<code>SWITCH_IDLE_EXIT_S</code>	<code>20</code>	Seconds with no sip/puff before it returns to <code>SWITCH_EXIT_GROUP</code> by itself. <code>0</code> = never (then use the long puff).
<code>SWITCH_EXIT_PUFF_S</code>	<code>5.0</code>	Hold one puff this long to leave straight away.
<code>SWITCH_EXIT_GROUP</code>	<code>1</code>	Where you go when leaving (1 = Keyboard).
<code>SWITCH_COUNTDOWN_S</code>	<code>10</code>	(v1.21+) Seconds of on-screen countdown before it leaves, e.g. <code>MOUSE IN 5</code> . <code>0</code> = none.

Key names are listed in the **Keycode Reference** (`KEYCODE_REFERENCE.md` / the PDF). For example, a game controlled with the arrow keys could use `SWITCH_SIP_KEY = "LEFT_ARROW"` and `SWITCH_PUFF_KEY = "RIGHT_ARROW"` .

The validator (`Check my AeroMorse files`) checks these settings before you copy `config.py` to the device.

Tips

- **Watch the top line.** It says **[SWITCH]** while you're in Switch mode. A wireless display mirrors it, so you can put it where you can see it while playing.
 - **Resting mid-game** for less than 20 seconds keeps you in Switch mode. If your game has longer pauses, raise `SWITCH_IDLE_EXIT_S` (e.g. `60`).
 - **Keys never get stuck.** The key is let go when your sip/puff ends, and also whenever AeroMorse leaves Switch mode.
 - **The sensor settings still apply** — `THRESH_SIP` / `THRESH_PUFF` decide how hard a sip/puff must be to count, the same as for Morse.
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Troubleshooting

Problem	Why	Fix
It leaves Switch mode while I'm playing	A pause longer than <code>SWITCH_IDLE_EXIT_S</code> , or a 5-second puff	Raise <code>SWITCH_IDLE_EXIT_S</code> / <code>SWITCH_EXIT_PUFF_S</code>
I need the mouse between games	Switch mode has no Morse, so there is no code to press	Set <code>SWITCH_EXIT_GROUP = 2</code> so leaving lands in Mouse, and add a short Group 2 code for "group 7" in <code>morse_map.py</code> to come back. The countdown (<code>SWITCH_COUNTDOWN_S</code>) shows when it is about to leave
It won't leave Switch mode	You keep sipping/puffing within 20 s	Stop for 20 s, or hold one puff for 5 s
Nothing happens in the game	The game window isn't in front, or wants other keys	Click the game once; or change <code>SWITCH_SIP_KEY</code> / <code>SWITCH_PUFF_KEY</code>
<code>...-----</code> doesn't go to Switch mode	<code>SWITCH_GROUP</code> isn't 7	Check <code>config.py</code> (or use the Group 0 code for the group you chose)
Group 7's own patterns don't type	Group 7 is Switch mode	Set <code>SWITCH_GROUP = 0</code> to use Group 7 for Morse again